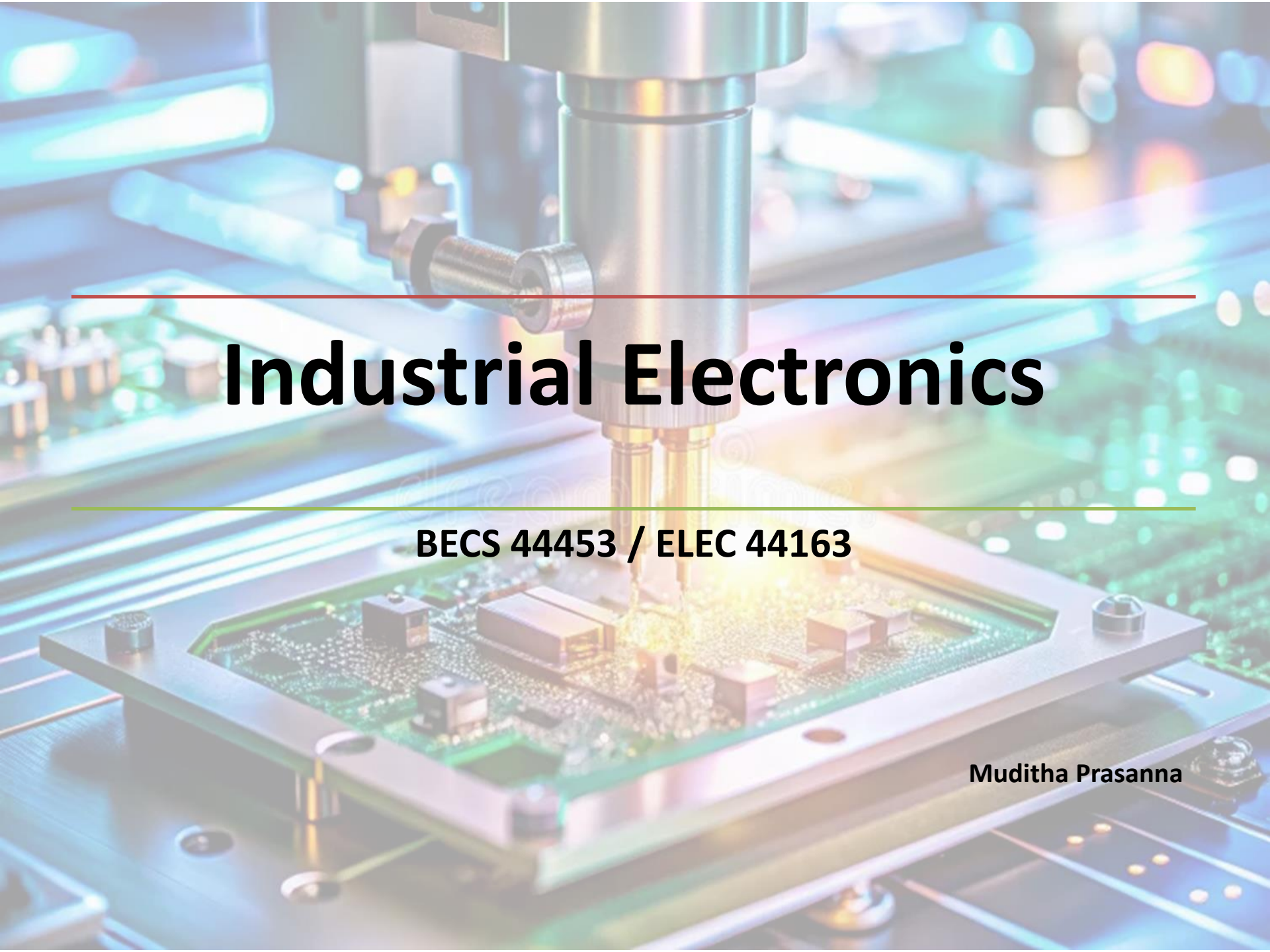
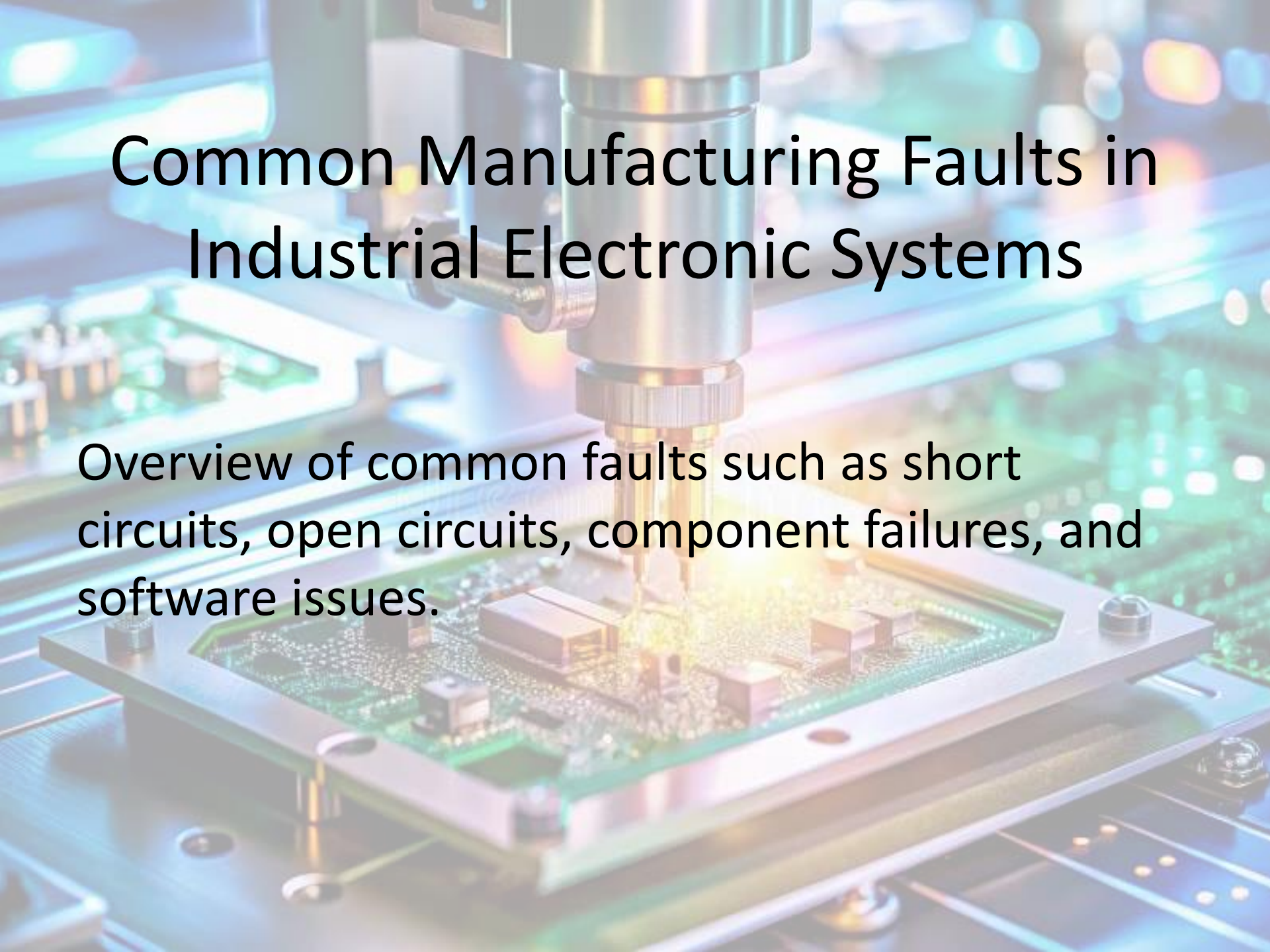


A close-up photograph of a precision industrial machine, likely a CNC mill or laser cutter, working on a green printed circuit board (PCB). The machine's tool head is positioned directly over the PCB, which is held in a metal fixture. The background is blurred, showing other parts of the factory floor with various lights and equipment. The overall color palette is dominated by the green of the PCB and the metallic tones of the machinery, with some blue and yellow highlights from the ambient lighting.

Industrial Electronics

BECS 44453 / ELEC 44163

Muditha Prasanna



Common Manufacturing Faults in Industrial Electronic Systems

Overview of common faults such as short circuits, open circuits, component failures, and software issues.

In-Circuit Testing (ICT)

- Definition and Purpose
- How ICT Works
- Advantages and Limitations

What is ICT?

- **In-Circuit Test (ICT)** is a method used to check the functionality of electronic circuit boards.
- Performed after manufacturing to detect manufacturing defects.
- Tests individual components and their connections directly on the PCB.

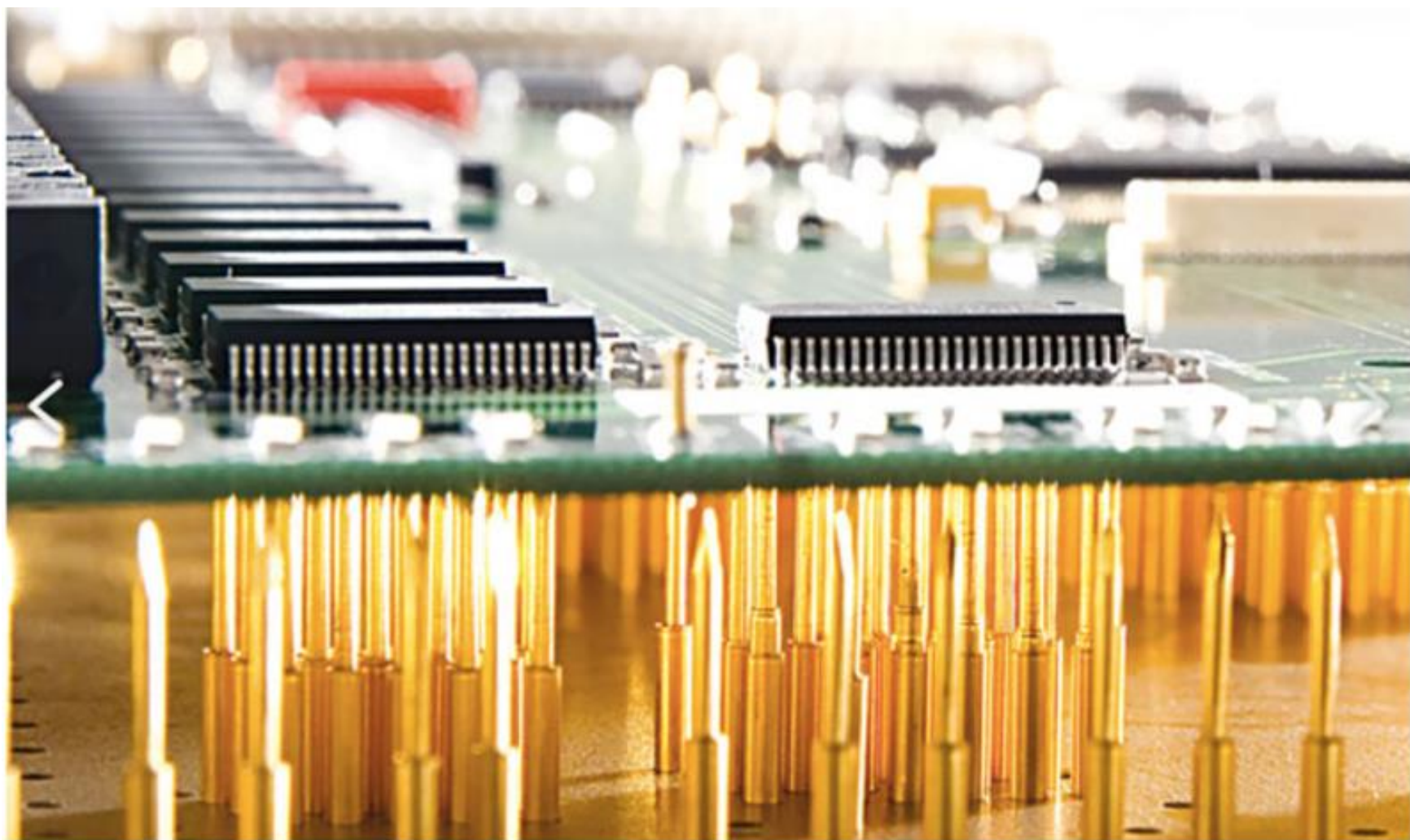
Purpose of ICT

- Detect faulty, missing, or incorrectly placed components.
- Verify electrical connections.
- Ensure the circuit meets design specifications.
- Reduce the risk of defective products reaching customers.

How ICT Works – Overview

How ICT Works – Overview

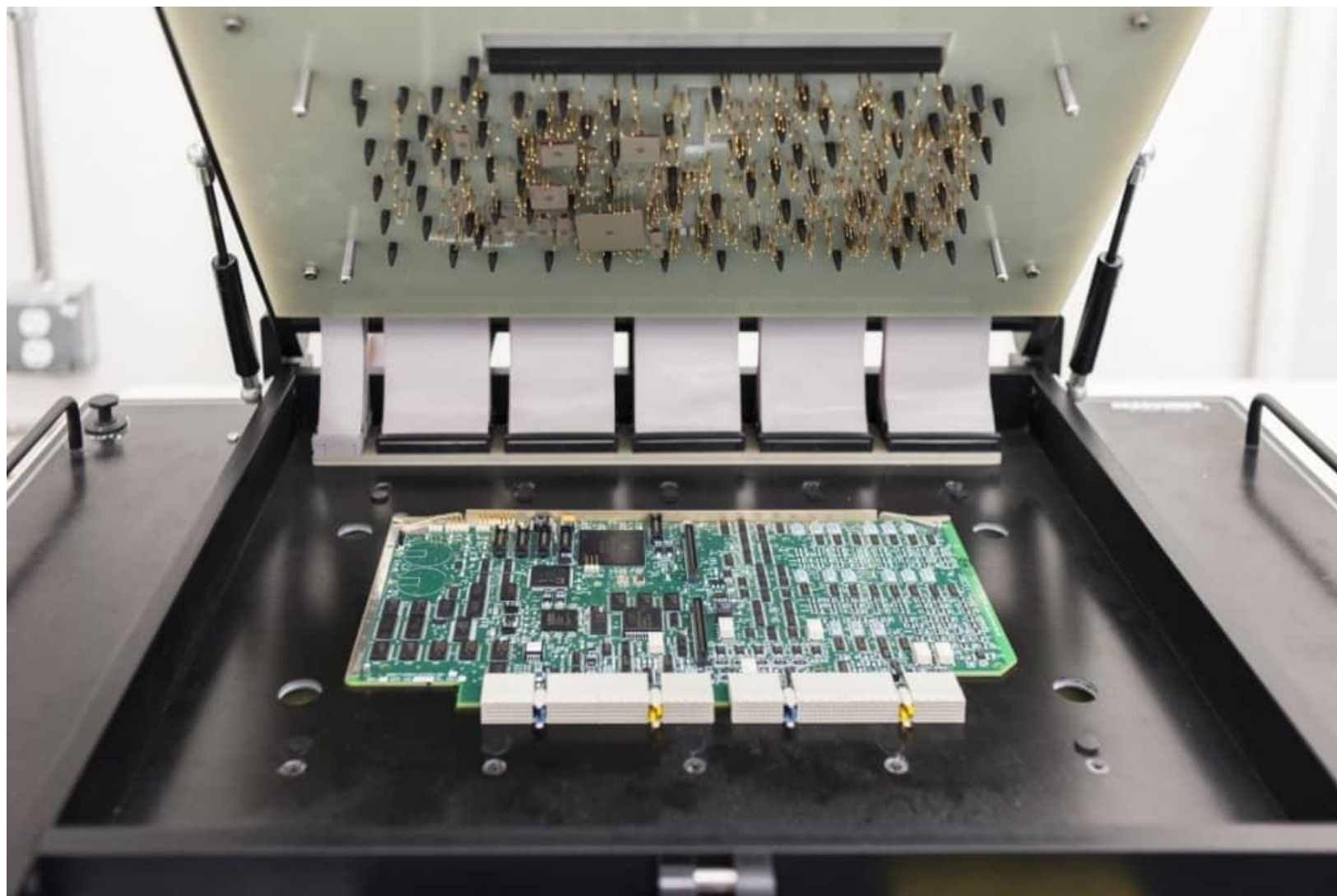
- Uses a bed-of-nails or fixture to connect test probes to various points on the PCB.
- Sends predefined test signals to check component responses.
- Measures electrical parameters to verify circuit integrity.



Components of an ICT System

Components of an ICT System

- **Test Fixture:** Custom fixture with probes that contact the PCB.
- **Test Equipment:** Provides test signals and measures responses.
- **Test Program:** Defines the test procedures and criteria.
- **Computer/Controller:** Manages testing and data collection



The ICT Process

- 1. Loading the PCB:** The board is placed into the test fixture.
- 2. Applying Test Signals:** The system injects signals into specific points.
- 3. Measurement & Analysis:** Responses are measured and compared to expected values.
- 4. Results Output:** Pass or fail is determined based on measurements.
- 5. Reporting:** Data is recorded for quality control.

Types of Tests in ICT

- **Continuity Testing:** Checks for open circuits.
- **Resistance Measurement:** Verifies resistor values.
- **Capacitance & Inductance Testing:** Checks passive components.
- **Component Presence & Placement:** Ensures all components are correctly installed.
- **Functionality Checks:** Some tests simulate operation to verify responses.

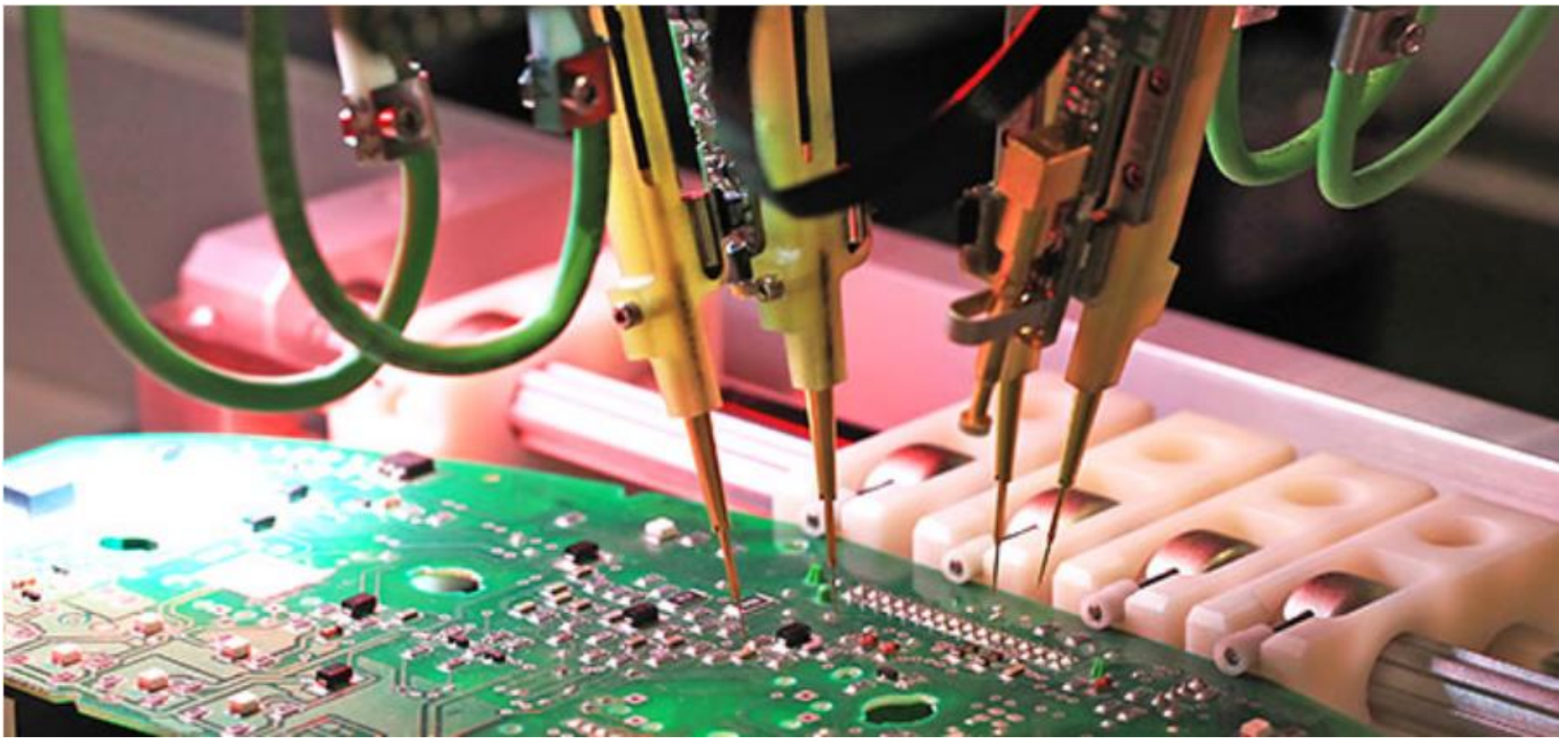
Limitations of ICT

- Requires custom fixtures for each PCB design.
- Limited to testing components and connections, not functional operation.
- Not suitable for very complex or multi-layer PCBs without advanced setups.
- Upfront cost for fixtures and setup.

Flying Probe Testing Overview

Flying Probe Testing Overview

- Uses movable, computer-controlled probes to test PCB electrical connections
- No need for custom fixtures, making it flexible and adaptable



Key Features & Advantages:

- Rapid setup & easy to modify test programs
- Cost-effective for small to medium production runs
- Suitable for complex and multi-layer PCBs
- Reduces lead times and hardware investment

Applications:

- Prototype validation
- Small batch manufacturing
- Repair and rework testing

Limitations:

- Not ideal for high-volume testing
- Slower compared to fixture-based methods

Applications of ICT

- Electronics manufacturing industry.
- Consumer electronics.
- Automotive electronics.
- Medical devices.
- Aerospace components.

Functional Testing



Functional Testing

- Definition and Purpose
- How Functional Testing Works
- Advantages and Limitations



PCBA Functional Testing involves verifying that the assembled printed circuit board **functions correctly** according to design **specifications**. It tests the electrical performance of the board in a simulated real-world environment.

Purpose:

- Detect faulty components or connections
- Ensure the PCB performs all required functions
- Prevent defective products from reaching customers
- Validate assembly quality and reliability

How PCBA Functional Testing Works

1. Test Preparation:

Set up testing environment with necessary fixtures and test equipment

2. Test Program Development:

Create test scripts and procedures based on PCB specifications

3. Test Execution:

Connect the PCB to the test system and run functional tests, simulating real operation

4. Data Collection & Analysis:

Record measurements, signals, and responses to verify performance

5. Defect Identification:

Identify issues such as open circuits, shorts, or faulty components

6. Report & Retest:

Document results, repair defects, and repeat tests if necessary

Advantages of PCBA Functional Testing

- Detects manufacturing defects early
- Ensures the PCB performs required functions before shipment
- Reduces field failures and returns
- Validates overall product quality and reliability
- Supports compliance with industry standards

Limitations of PCBA Functional Testing

- Requires specialized testing fixtures and equipment
- Can be time-consuming and costly for complex boards
- May not detect all component-level faults (e.g., subtle component degradation)
- Needs accurate test programming and calibration
- Not a substitute for other types of testing like in-circuit or environmental testing

Advanced Troubleshooting Techniques

- Using Oscilloscopes
- Using Multimeters
- Using Logic Analyzers

PCB Assembly Testing

- Visual Inspection of PCBAs
- Automated Optical Inspection (AOI)
- X-ray Inspection

Visual Inspection of PCBAs

Table 1-2 Inspection Magnification (Land Width)

Land Widths or Land Diameters ¹	Magnification Power	
	Inspection Range	Maximum Referee
> 1 mm [0.04 in]	1.5X to 3X	4X
> 0.5 to ≤ 1 mm [0.02 to 0.04 in]	3X to 7.5X	10X
≥ 0.25 to ≤ 0.5 mm [0.01 to 0.02 in]	7.5X to 10X	20X
< 0.25 mm [0.01 in]	20X	40X

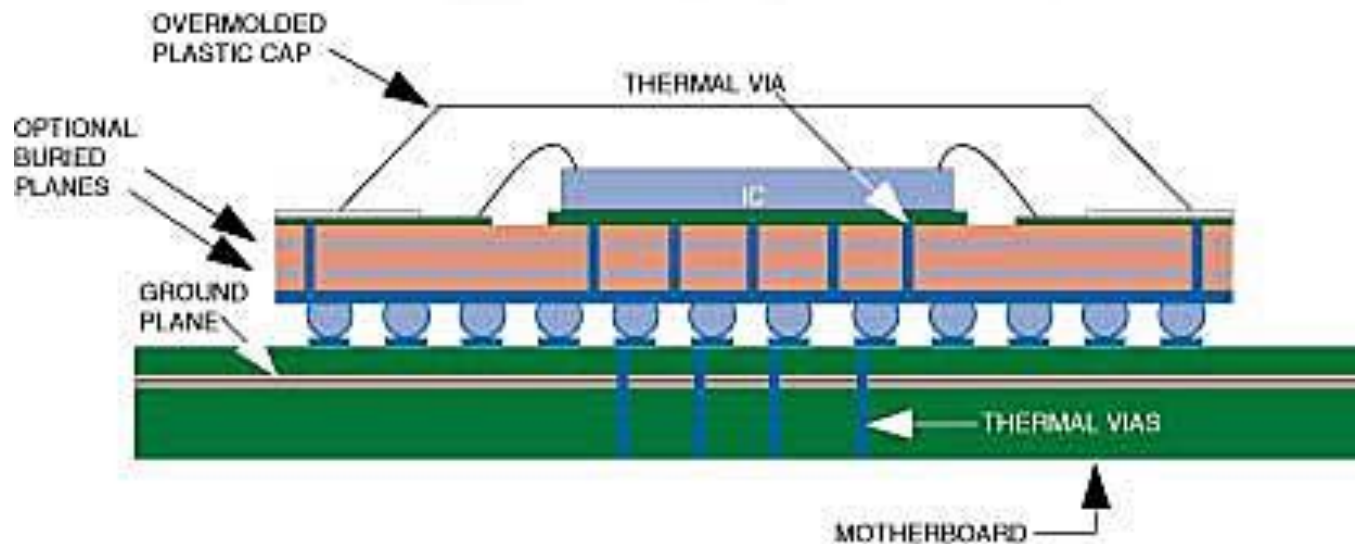
Table 1-3 Magnification Aid Applications For Wires And Soldered Conductors

Wire Size AWG Diameter mm [inch]	Magnification Power	
	Inspection Range	Maximum Referee
Larger than 14 AWG > 1.63 mm [> 0.064 in]	N/A	1.75X
14 to 22 AWG 1.63 – 0.64 mm [0.064 to 0.025 in]	1.5X – 3X	4X
< 22 to 28 AWG < 0.64 mm – 0.32 mm [< 0.025 – 0.013 in]	3X – 7.5X	10X
Smaller than 28 AWG < 0.32 mm [< 0.013 in]	10X	20X

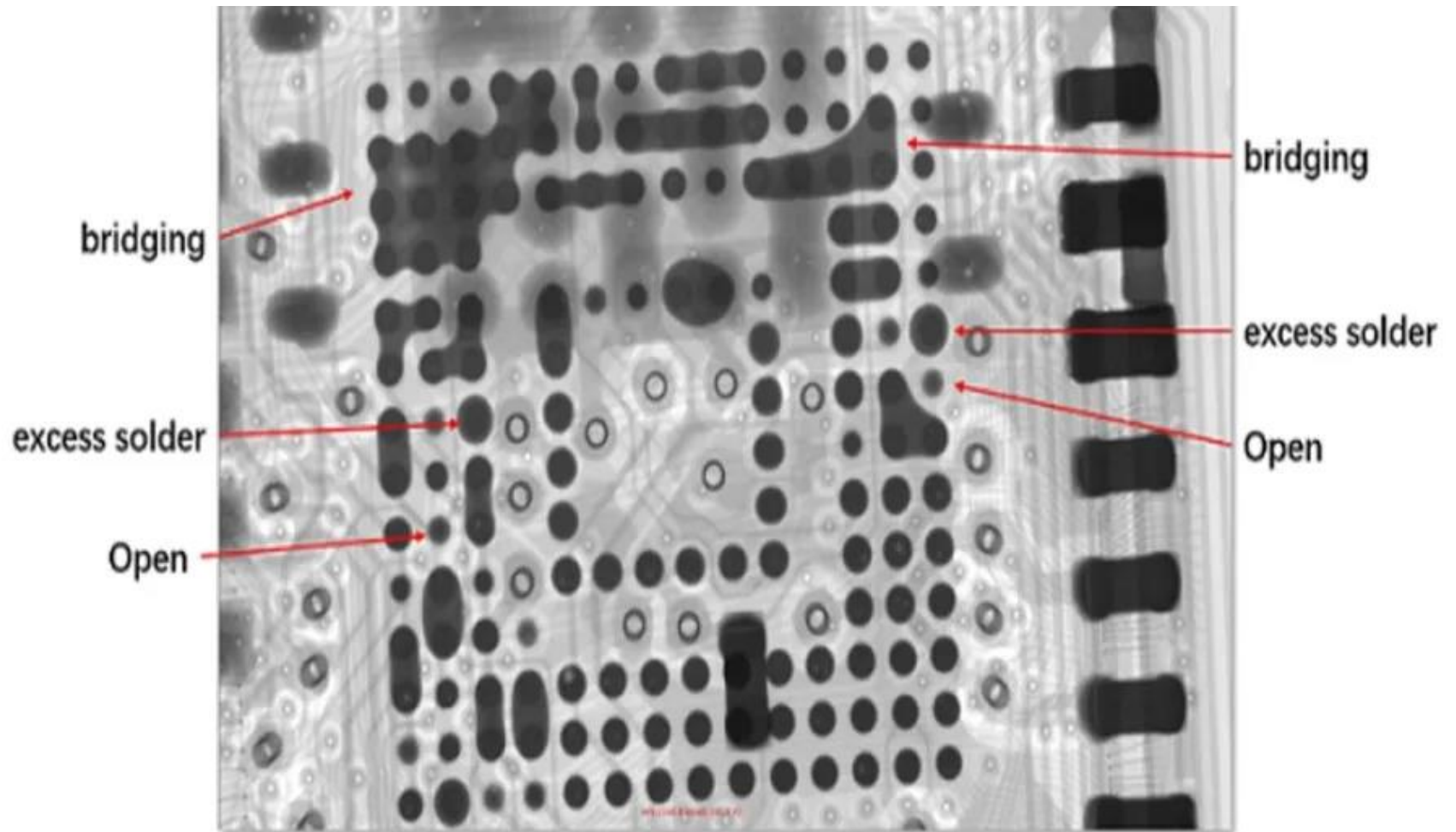
Automated Optical Inspection (AOI)



X-ray Inspection



X-ray Inspection



PCB BGA, NG

Environmental Testing

- Temperature Testing
- Humidity Testing
- Vibration Testing



Software Tools for Troubleshooting

- Diagnostic Software
- Simulation Software
- Firmware Updates

Documentation and Record Keeping

- Importance of Documentation
- Types of Records to Keep
- Best Practices

Training and Skill Development

- Importance of Training
- Training Methods
- Continuous Learning

Common Issues and Solutions

- Power Supply Issues
- Signal Integrity Issues
- Component Failures

Repairing Faults

- Step-by-step guide to repairing faults, including component replacement, soldering techniques, and software updates.

Preventive Maintenance

- Importance of preventive maintenance and techniques to ensure the longevity of electronic systems.

Scheduled Maintenance

- Creating and implementing a scheduled maintenance plan for industrial electronic systems.

Predictive Maintenance

- Using data analysis and machine learning to predict and prevent potential failures.

Maintenance Tools and Equipment

- Overview of tools and equipment used for maintenance, including multimeters, oscilloscopes, and diagnostic software.

Safety Precautions

- Safety measures to be taken while troubleshooting, repairing, and maintaining electronic systems.

Documentation and Record Keeping

- Importance of documentation and record keeping in maintenance activities.

Advanced Troubleshooting Techniques

- Introduction to advanced troubleshooting techniques such as thermal imaging and signal analysis.

Component Testing

- Methods to test individual components such as resistors, capacitors, and integrated circuits.

Software Troubleshooting

- Techniques to troubleshoot software issues in industrial electronic systems.

Network Troubleshooting

- Identifying and resolving network issues in industrial electronic systems.

Power Supply Troubleshooting

- Troubleshooting power supply issues and ensuring stable power delivery.

Environmental Factors

- Impact of environmental factors such as temperature, humidity, and dust on electronic systems.

Training and Skill Development

- Importance of training and skill development for maintenance personnel.

Industry Standards and Regulations

- Overview of industry standards and regulations related to electronic systems maintenance.
 - IPC A 610
 - IPC WHMA 620
 - IPC 7711/21

Emerging Technologies

- Introduction to emerging technologies in industrial electronic systems.
 - AI, IoT, and 5G are rapidly transforming industrial electronic systems, leading to increased automation, efficiency, and data-driven decision-making.

IoT in Industrial Systems

– **Connectivity and Data Collection:**

- IoT devices, equipped with sensors, are connecting industrial equipment and machinery, enabling real-time data collection and analysis.

– **Remote Monitoring and Control:**

- IoT enables remote monitoring and control of industrial processes, allowing for greater flexibility and responsiveness.

– **Enhanced Efficiency:**

- Data from IoT devices can be used to optimize energy consumption, reduce waste, and improve overall operational efficiency.

Cybersecurity in Industrial Systems

- Importance of cybersecurity and techniques to protect electronic systems from cyber threats.

Remote Monitoring and Maintenance

- Using remote monitoring and **maintenance** techniques to ensure system reliability.

Energy Efficiency

- Techniques to improve energy efficiency in industrial electronic systems.

Sustainability in Maintenance

- Implementing sustainable practices in maintenance activities.

Future Trends in Industrial Electronic Systems

- Predictions and trends for the future of industrial electronic systems.